SHARP Manual Lathe

Standard Operating Procedure

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1. Introduction

This document is to be used by properly trained and experienced students and staff as a guideline for the safe operation of this machine and is written in accordance with OSHA and PERRP standards.

2. Scope and Application

This manual lathe (knee mill) is to be used only for material removal and measurement purposes in service to the Engineering Student Projects teams housed in this facility, as well as for support of said facility. Students and Staff who wish to use this equipment must have first received training from the Shop Supervisor and have read and signed both the Motorsports Disclosure Form and the Machine Shop Access Form. This is described in greater detail in SECTION 6.

3. Responsibilities

- The Shop Supervisor will update and ensure the availability of this document at the machine.
- The Shop Supervisor (and staff when instructed) will solely be responsible for performing/arranging maintenance, repair and machine modifications to ensure the machine is in safe operation condition.
- Student Operator is required to have read, signed and turned in the appropriate forms and have participated in task specific training instructed by the Shop Supervisor before operating any equipment in the Facility. This is described in greater detail in Section 6.
- Student Operator is responsible for all cleaning and housekeeping. This machine must be kept in showroom condition as it is on display and represents both the Student Operators and the University. Leaving the machine in an unclean state because the previous operator did not perform his/her housekeeping duties is not acceptable. Restating, at all times the following rule applies; when the Student Operator leaves the machine, it is to be left in a safe, clean condition.
- Student Operator is responsible for using the machine's safety equipment (provided by the Staff) as well as PPE (provided by the Student Operator's Team) in a manner that is both effective and in accordance with OSHA, PERRP, and industry standard, where applicable. Failure to employ required safety measures because of ignorance of laws, rules and standards is not acceptable and grounds for expulsion from the Facility. When in doubt, ask!
- Upon discovery of any missing guards, damaged equipment or any other form of safety hazards, the Student Operator must immediately cease operation and inform the Shop Supervisor, regardless of fault or involvement.
- Student Operator is responsible for notifying the Shop Supervisor of any crash or accident occurring during operation. Internal damage may not be visible and presents a hidden hazard to other users; the Shop Supervisor must be made aware so that potential problems can be sought out and cumulative wear/damage tracked.

- Student Operator is responsible for properly shutting down and returning the machine to a ready condition for the next operator. Typically this is defined as returning the machine to the state in which you found it, albeit cleaner, with associated tools in their proper locations as specified in SECTION 4k.

4. Standard Operation of the Equipment

- **Power Switch, Foot Brake and Engagement Lever**
- **3-Jaw, 4-Jaw and Collet Chuck – Clamping and Operation**
- **Speed Selection**
- **Carriage, Cross Slide and Compound Movement**
- **Operating the Tailstock**
- **Drilling**
- **Turning, Facing, Threading and Parting**
- **Power Feed**
- **Using Dials / Digital Readout**
- **Required Maintenance – Pre and Post Inspection**

  BEFORE operating the machine, the Student Operator is responsible for the following;
  - Ensure that the floor mats are swept clean and lay flat on the floor without creating a trip hazard or impeding access to safety features of the machine.
  - Be sure to remove tools, parts, swarf/chips, etc. that previous operators may have left on the machine and that operation of any feature of the machine will not be impeded by objects sitting nearby.
  - The axis guides/areas of sliding contact of the machine must be gently BRUSHED clean. Do this in a way that moves chips away from sealing surfaces and DO NOT use compressed air to clean the exposed axis guides/dovetails/linear slides as this tends to wedge chips/dust/swarf deeper into the sealed regions of the machine.
  - If the machine is equipped with automatic oiling system(s), check that the fluid levels in the lubrication reservoirs are acceptable.
  - If the machine is equipped with manual or spring-return oiling systems, make sure to actuate those prior to moving the machine to prevent metal-on-metal contact in the sliding portions (“ways”) of the machine.

  AFTER operating the machine, the Student Operator is responsible for the following;

  - Ensure the tools necessary for the machine are back in their proper locations.
  - Clean the machine and area of ALL swarf/chips.
  - Make sure to clean under the floor mats as well as sweep the swarf/chips from the surface and that the floor mats lay flat without creating a tripping hazard.

- Ensure that machine guards are in place and in good condition for the next operator.

5. Personal Protective Equipment

a. Required PPE

- Clear eye protection, in good condition, must be worn at all times in the Facility to guard against flying debris. In situations where shaded eye protection (goggle or glasses) is necessary to avoid exposure to radiation sources, clear eye protection may be supplemented with the appropriate shaded eye protection, however, clear eye protection must still be worn under welding helmets.
- Face shields, fire-retardant aprons and gloves may be necessary in certain operations producing hot or flying swarf/chips. However, wearing gloves may create additional hazards around pull-in and pinch points. See the Shop Supervisor for advice on appropriate PPE in special cases.
- Closed toed shoes and long pants must be worn at all times in the Facility.
- Clothes having a high cotton content are highly recommended. Many athletic/sport oriented synthetic clothing has a low melting temperature and can cause serious burns to the operator's skin if exposed to heat sources like hot workpieces, cutting tools and swarf/chips.

b. PPE Storage

- The teams in the Facility are individually responsible for ensuring the availability of PPE for the Student Operators.
- Specialty PPE for one-time operations may be requested/borrowed. See the Shop Supervisor in these cases.

c. PPE Inspection

- The teams are required to ensure that, at all times, appropriate PPE is available for the operations the team regularly performs.
- The teams must keep enough quantity of the appropriate PPE to ensure that all members participating in a given operation have PPE available to them.

6. Required Training

All operators of the manual lathes located at the Facility are REQUIRED to have documented training. Documentation must include facility activity disclosure forms as well as machine training. The training must be conducted by the Shop Supervisor and be on file at the Facility BEFORE the operator is permitted access to the machine. The training will cover the operation of the manual milling machine as well as, including but not limited to, the following topics;

- Power Switch, Foot Brake and Engagement Lever
- 3-Jaw, 4-Jaw and Collet Chuck – Clamping and Operation
- Speed Selection
- Carriage, Cross Slide and Compound Movement
- Operating the Tailstock
- Drilling
- Turning, Facing, Threading and Parting
- Power Feed
- Using Dials / Digital Readout
- Required Maintenance
- Understanding of the SOP that is written
- Understanding of the Job Hazard Analysis / Required PPE
- Common causes or misuse that can lead to damage or injury
- Pre-Inspection and what to do if there is a safety issue or service issue needed for the equipment
- Emergency Procedures

7. Emergency Procedures

a. Equipment Malfunction/Tool Breakage/Fixturing Failure

Upon any failure of the equipment, tool, or fixturing, immediately remove the tool from contact with the part and turn the machine off using the method described in SECTION 4 - Power Switch, Foot Brake and Engagement Lever. Removal of tool contact is usually possible by retracting the Cross Slide, as described in SECTION 4 – Carriage, Cross Slide and Compound Movement. Before turning the machine on again, check to ensure the following;

- Check that the Tool is clamped securely and is not damaged from the incident
- Check that the Part is clamped securely: If the part becoming loose was the reason for the stoppage, DO NOT PROCEED by re-fixturing in the same manner. Contact the Shop Supervisor for advice or seek assistance from more experienced Student Operators.
- Check that the machine is safe to turn back on: If the machine did not perform as expected, i.e. spindle rotation reversed suddenly, spindle rotation stopped or any handle required higher than usual force input, contact the Shop Supervisor before proceeding.

b. Building Evacuation/Announced Emergency

Cease operations immediately by removing tool from contact and turning off the power to the machine as described in SECTION 4 - Power Switch, Foot Brake and Engagement Lever and SECTION 4 – Carriage, Cross Slide and Compound Movement. Ensure that any items capable of generating immediate danger, i.e. flammables, sharp protruding objects, etc., are taken care of as best as possible in an expedient manner and calmly exit the building or follow the directions of the most senior person in the Facility.

c. Power Outage/Unexpected Emergency

During a power outage, all equipment should be turned off to prevent mechanical and electrical hazards to the Student Operators in the building. The equipment should be confirmed to be switched off so that, in the event of a power outage and subsequent power restoration, the equipment does not suddenly and unexpectedly start up. During heavy storms, the operations should be ceased and the equipment should be powered off for the duration of the storm so that the operator doesn't experience an unexpected stoppage that could result in tool breakage and personal injury.

8. Lock Out/Tag Out

This manual lathe is equipped with a lockout compatible power disconnect switch mounted on the wall behind the machine. If color-coded lock-out equipment is present on the disconnect switch, preventing the powering of the machine, the machine is to be considered removed for operation until the person who installed the lock-out gear removes it after confirmation of completed repairs to the machine. DO NOT attempt to remove the lock-out equipment or alternatively power the machine until the machine is reintroduced to service by the lock-out tag-out authorized personnel.

As of the following date, the people listed below are the sole authorized personnel allowed to perform lock-out tag-out operations. This includes the act of locking out to remove from service as well as the act of lock removal and reintroduction to service.

As of 1-1-17, the following are authorized lock-out tag-out personnel;

- Darrin Orr
- Matthew Little
- Don Williams

Any Student Operator who has reason to believe a machine needs service for any reason, safety or otherwise, is responsible for contacting the Shop Supervisor immediately.

Appendix A

Figure 1

Figure 2

Figure 3

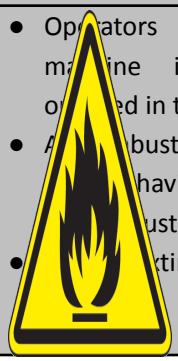
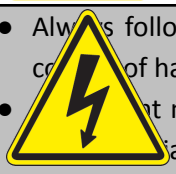
Figure 4

Figure 5

Figure 6

Figure 7

Appendix B - Job Hazard Analysis

Work Duties	Potential Hazards	Controls
Operating the Machine	**Flying Debris**	 • Saf[illegible] glasses are required while o • M[illegible]e mounted guards must b[illegible] [illegible]riate manner
Operating the Machine	**Pinch, Crush and Entanglement Hazards**	 • Ref[illegible] to SOP and training fo[illegible] pr[illegible]dures. • U[illegible]aution while handling heavy c[illegible]s
Operating the Machine	**Entanglement, Lacerations and Flying Debris Hazards**	• During Tool Changes, machine mu • Operator must be trained in th[illegible] use of the brake to prevent [illegible] startup • Never reach towards the [illegible] operation while machine is runn
Operating the Machine	**Fire**	 • Op[illegible]ators are trained to shut d[illegible] ma[illegible]ine in accordance with [illegible] o[illegible]d in the machine-specific SOP • A[illegible]bustibles must be clear of th[illegible] [illegible]havings, combustible dust [illegible]ust be cleaned up appropria[illegible] • [illegible]xtinguishers must be acces[illegible]
Operating the Machine	**Slips, Trips and Falls**	 • Proper housekeeping must be m[illegible] at all times, including cleaning [illegible] immediately
Machine Service	**Electrocution/Accidental Start up**	 • Alw[illegible]s follow Lock out / Tag out c[illegible] [illegible]of hazardous energy • [illegible]t needing serviced shall [illegible]ately by authorized per[illegible] • Service should be completed in [illegible] the manufacturer requirements b[illegible] and authorized personnel or a cont[illegible]

Post-Operation Housekeeping	**Lacerations or Foreign Body in Eyes**	 ● W[illegible]n cleaning up safety glasses a[illegible]e worn at all times ● [illegible]ttempt cleaning up swarf/c[illegible]s running ● Gloves should be used with cautio machinery, as they may pre entanglement hazard